

Investigating the trade-offs between accuracy and transferability for ASAG models using experiments for direct comparison

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ABSTRACT

Automated short answer grading (ASAG) has the potential to significantly impact modern pedagogic techniques, yet the identification of optimal model-building approaches is challenging due to varied datasets and inconsistent performance metrics. This study addresses these challenges by utilising an array of public datasets and evaluating various models' performance using diverse metrics both within individual datasets and across different domains and datasets. The adaptability and performance of various models, including interpretable models with separate 'embedding' and 'grading' processes, the state-of-the-art BERT model, and a Large Language Model (LLM), are examined. This rigorous approach enables direct model comparisons and provides critical insights into their performance. The study contributes significantly to the ASAG field by offering an in-depth analysis of model performance within and across different domains and underscores the promising potential of LLMs for ASAG. The BERT model achieves the highest performance on specific datasets, while the LLM model (GPT-3.5Turbo) obtains the most consistent performance across datasets and domains.

KEYWORDS

ASAG, BERT, LLM, Cross-domain, Cross-dataset

GITHUB

The code for this thesis can be found at the following Git repository.

1 INTRODUCTION

An imperative part of the learning process is assessing the knowledge of pupils. An accurate understanding of pupils' knowledge makes learning more personalized and can make measuring the effectiveness of different pedagogic techniques easier. The primary method to assess students' knowledge is through summative assessment (written examinations) made up of different types of questions such as closed-form questions (multiple-choice) or open-answer questions (essay or short answers) [11]. This paper focuses on the short form of the open-answer questions. Graders assess the answers to these questions based on their similar meaning, not on their being identical. This requires graders to actively think while grading, making the grading process time-consuming. "Automatic Short Answer Scoring (ASAG) refers to the scoring of answers to short answers without human intervention" [26] and aims to make this valuable pedagogical asset less labour intensive. Short answer questions are defined by the following criteria [6]:

- The answer to the question cannot be found from within the question and requires external knowledge to be answered.
- The answer must be given in natural language.
- The answer length should roughly be between one sentence and one paragraph.
- Graders should assess the answers based on content rather than writing style.

ASAG faces significant challenges such as scarcity of training data, diversity of domains, the need for model interpretability, high density of content in the texts, and considerable variability in content expression [32]. Despite these challenges, significant developments in natural language processing over the past fifteen years have enhanced the performance of ASAG models [4, 6, 11].

Emerging out of these advances in NLP, the models in Automated Short Answer Grading (ASAG) have evolved to broadly fall into two categories: those that separate the processes of embedding and grading and those that integrate these functions into a single model. The former approach, referred to as 'embedding-based models' in this article are relatively interpretable as the parts of the student answer that move the answer further away from the reference answer can be visualized [1, 34]. Models that integrate both embedding and grading tasks achieve the current state-of-the-art ASAG performance, specifically those using pre-trained Bidirectional Encoder Representations from Transformers (BERT). Remarkably, these models outperform (on accuracy and weighted F1) human graders on two, not publicly available, psychology datasets when evaluated through majority voting (based on the human graders) [32]. Generative Pretrained Transformer (GPT) models have shown robust performance across various NLP tasks [9]. However, their application, specifically as grading models is relatively unexplored, with one study employing GPT and GPT-2 only for embedding [9]. Furthermore, Large language models (LLMs), such as GPT-3 and variants like GPT-3.5-turbo, have demonstrated strong performance in NLP tasks through few-shot learning [5]. Despite this, their application in ASAG research is yet to be explored. A detailed exploration of the evolution of the ASAG field can be found in the related work section (2).

Comparisons between studies on ASAG are often challenging. Research frequently relies on unique datasets that are typically not publicly available. Notable exceptions include the Beetle, Texas, ASAP-sas, and ScientsBank datasets. These studies employ a diverse range of metrics—from RMSE and correlation to Quadratic Weighted Kappa (QWK) to accuracy and weighted F1-score. This diversity, combined with differing dataset difficulty levels, creates challenges in cross-comparing model performances across datasets and domains [6, 11].

This paper aims to fill three gaps in the current literature: (1) the lack of performance measurements on diverse datasets of the BERT model, (2) the limited out-right performance comparisons between more interpretable models where embedding and grading are separated and models where they are not and (3) the absence of research on the effectiveness of LLM models as grading models. The related research question and sub-questions are:

What are the trade-offs observed between traditional ASAG models (where embedding and grading is split), the current state-of-the-art ASAG model (BERT) and large language model (LLM) GPT-3.5turbo in performance on the same datasets and what is their impact on transferability to new datasets and domains?

- To what extent do grading models, where embedding and grading are separated, maintain their performance across datasets on new datasets and new domains?

- To what extent does the current state-of-the-art ASAG model, a BERT model, maintain its performance across datasets on new datasets and new domains?
- How do LLM models (GPT-3.5turbo), using zero, three and four shot learning, perform across datasets, new datasets and new domains?
- How do models where embedding and grading are separated (embedding-based models), BERT and LLMs compare in their performance across datasets, on new datasets and new domains?

This study uses a large variety of publicly available datasets and tests the performance of diverse models on each dataset individually. To measure generalization, all the datasets are also concatenated, and cross-validation is conducted with one dataset dropping out for each fold. Finally, the data inside all the datasets of which the domain is known are concatenated, and the same cross-validation is done on the domains.

Three main types of approaches are tested. The first approach, representing more interpretable approaches, tests various models to embed the words in the student and reference answer and trains regressions on the resulting cosine similarity. This approach is known to be more interpretable than the pre-trained models and indicates the performance of embedding-based models across new domains. The second approach trains BERT for five and ten epochs on a sample of 2,000 student answers for each experiment. The third and final approach tests the accuracy of an LLM model (GPT-3.5turbo) using zero, three and five-shot learning. The examples (shots) are added in accordance with the data splits tested.

The study finds that BERT exhibits superior performance to the LLM, particularly on smaller datasets, reflecting its ability to identify grading attributes within specific contexts, potentially at the risk of overfitting. However, the LLM shows more consistently strong performance across varied datasets and domains, highlighting its robustness and generalization capabilities.

2 RELATED WORK

2.1 Embedding-based models

Early machine learning approaches in automated short answer grading (ASAG) employ a range of natural language processing (NLP) techniques to generate various representations of student and reference answers. Each student answer representation is compared to its corresponding reference, with the results of these comparisons transformed into features (e.g., the percentage of identical tokens). Machine learning models then use these features to predict the number of points to assign to the student answer [2, 11, 12, 17, 23]. These methodologies tend to abstract away variations in sentence form, allowing sentences with the same lexical items but different syntactic relationships to have similar semantic interpretations [10].

To capture more semantic features in the text, corpus-based approaches have been developed, which aim to learn relevant features of words by considering the context in which they appear. These techniques, such as tf-idf, LSA and ESA [15, 22], are used to create numerical representations of the student answers and reference answers. These representations quantify the semantic similarity

between student answers and reference answers, which in turn is used to assign the points.

The accuracy of word representations inside vector space improved substantially with the advent of new word embedding methods [18, 19, 39]. The word embeddings created by models such as Word2Vec [18], Fasttext [3], and Glove [25] seem to be able to learn a multitude of relevant features and embed these into the different dimensions of the vector representation of each word. Consequently, words with similar meanings have similar locations in this vector space. Cosine similarity, initially used on continuous bag-of-words (CBOW) model ranking features [20], has been found to perform well at determining the distance of words represented in these embedding spaces and is widely adopted in the literature [9, 18, 20, 39]. The word and sentence embeddings created by these models and various features representing their distance are used to increase the performance in divergent text classification applications [16, 38] and have been found to slightly increase the accuracy of ASAG [16].

Later approaches use neural networks to interpret the text embeddings. These studies train neural networks on the embeddings produced by Word2Vec, Glove and Fasttext and achieve high accuracy on text classification [8, 13, 36]. In one study, the embedded words are fed into a convolutional neural network (CNN) with three convolution, max pooling and dropout layers and one fully connected neural network to classify a new story into its topic with up to 97% accuracy [8].

This approach of training a neural network model on text embeddings is also adopted in the field of ASAG. By using both BLSTM and CNN models followed by a hybrid attention layer and softmax to grade the student answers, an F1 score of 0.9694 on a Chinese dataset after 100 training epochs is achieved [26], though more traditional embedding methods are used. A similar approach is adopted on the ASAP-sas dataset (also used in this study) using a CNN layer to interpret the embeddings, followed by LSTM, attention layer, and finally, a fully connected layer to make the prediction [30]. On the ASAP-sas dataset, this approach achieves 0.7439 mean Quadratic Weighted Kappa (QWK). A different paper using the same dataset expands this model to be more sensitive to the presents of the key elements of the reference answer inside the student answer. A word-level attention mechanism is used to identify the presents of these key elements. The vectors produced by the original model and the combined vectors from the attention mechanism are combined using the sigmoid function. Finally, a regression is trained to grade the student answer [34]. Attempts also aim to make these LSTM models more interpretable by visualizing what words brought the answer closer to the reference answer [1, 34]. Though these embedding models perform well on within-domain datasets, there is some indication that by combining learned and hand-crafted features, model performance can be improved on cross-domain ASAG datasets [31].

The deep learning methods discussed are often limited by the small size of the ASAG datasets available. Transfer learning aims to mitigate this problem by utilizing the knowledge embedded inside pre-trained models and then teaching these models a new task [7, 27, 37]. The models made for fine-tuning are trained on downstream tasks with a large text corpus. This article experiments

with two such models that are widely adopted: Bidirectional Encoder Representations from Transformers (BERT) and Generative Pretrained Transformer (GPT).

2.2 GPT

The initial GPT model is trained on a large corpus of unique unpublished books by learning to maximize the log-likelihood of correctly producing the following word given a sequence of words from the corpus [27]. The larger GPT-2 model is trained on a large corpus scraped from the internet [28]. The GPT-2 architecture is similar to the original GPT model, though GPT-2 is larger and has a larger context window (from 512 to 1024). The article tested the performance of GPT-2 on diverse tasks with zero-shot learning. It demonstrated that, though not achieving state-of-the-art performance, models trained on vast corpora "begin to learn how to perform a surprising amount of tasks without the need for explicit supervision" [28]. GPT-3, essentially a significantly enlarged version of GPT-2 (1.5 billion parameters) pushing GPT-3 (175 billion parameters) into the realm of LLMs, performs significantly better than its predecessor on similar tasks. The model is tested using few-shot learning, where the model is given a k amount of examples before making a prediction, rather than being fine-tuned on a given task or dataset. This greatly reduces the need for extensive task-specific training data and computational power to fine-tune the model. The article compares the performance of GPT-3, using zero, one and few-shot learning, with that of a variety of fine-tuned BERT models and finds that GPT-3 achieves promising results, sometimes even outperforming state-of-the-art models fine-tuned on the specific task [5]. As far as I know, few-shot learning has not been applied within the ASAG field using the GPT-3 model.

2.3 BERT

Bidirectional Encoder Representations from Transformers (BERT), as opposed to GPT, employs a bidirectional training approach and uses two tasks during pre-training. In the first task, 15% of the tokens in the input is masked, and the model is trained to maximize the likelihood of predicting the correct masked tokens throughout the training data. The second task is designed to help the model learn the relationships between sentences. It replaces 50% of the subsequent sentences in the training set with a random sentence from the corpus and trains the model to accurately predict whether this following sentence is the original subsequent sentence or not. The pre-training data comprises the BooksCorpus [40] and a portion of the English Wikipedia. BERT, in both its base and large versions, outperforms the original GPT model on a variety of NLP tasks. Given the architectural similarities between GPT and BERT, this suggests that models allowing for bidirectional training may offer performance advantages on downstream tasks.

2.4 ASAG using BERT and GPT

Both GPT and BERT are capable of producing token embeddings. One study compares a large variety of traditional (BOW, tf-idf, Word2Vec, GloVe, FastText) and contemporary (BERT, GPT, GPT-2) models to embed the student and reference answers. By using the cosine similarity of the aggregated token embeddings and a subsequent regression model, the student answers are graded. The

article finds marginal differences between the performance of the models, demonstrating that more traditional embeddings often even outperform the state-of-the-art embedding models. The paper argues that the marginal difference in performance could be related to the training data of the embedding models [9]. As far as I know, no studies have investigated whether these results remain the same if the models grade new domains. The knowledge obtained by the larger training corpora of GPT and BERT might give these models an edge when grading new domains. The approaches that split the embedding and grading phases do not achieve state-of-the-art performance, but by splitting up the embedding and grading phases, the grading system is more interpretable than models based entirely on BERT and GPT [1, 9, 11, 34].

A fine-tuned BERT model is observed to outperform individual graders on two psychology ASAG datasets, using majority voting between the three human graders as the reference grade. BERT achieves a 91.8 and 91.0 weighted F1 score on the two complete datasets and 88.1 and 90.0 when trained on 40% of the data [32]. The performance of the models, assessed on the test set after 4 and 12 epochs, does not exhibit substantial differences. However, the performance is considerably reduced if the training is conducted on one dataset and tested on the other, resulting in 70.0 and 75.5 weighted F1 scores (using 40% of the dataset). When the model is trained on both datasets, the performance on each dataset is comparable to the performance of the models trained solely on their respective datasets (again, using 40% of the data). The extent of the BERT model's performance on entirely new domains, however, remains to be seen. The study also reveals that BERT achieves a 75.8 weighted F1 score on the SemEval-2013 dataset. Slightly lower performance is documented on the same dataset in another article, a 72.8 weighted F1 score [26]. This paper also reports that RoBERTa, a BERT model pre-trained on a multilingual dataset, achieves a 1.7% higher weighted F1 score and that it retains its knowledge across languages.

3 METHODOLOGY

3.1 Data

This study will use a combination of multiple datasets to train and test the models created. All these datasets are publicly available online. Table 1 contains information on each dataset. The data descriptions can be found inside Table 1. More detailed descriptions of the datasets and the assigned points distribution for each dataset can be found in the appendix A.1.1.

This study processes the datasets in two phases: 'standardization and basic cleaning' and 'text data model preparation'. A detailed description of these phases can be found in the appendix: the former, where all datasets are standardized into a standard format, in Section A.1.2 and the latter, where the text processing is tailored to each model to best represent the pre-training data, in Section A.1.3. For each student answer, a normalized assigned grade between zero and one is calculated based on the maximum obtainable points of the related question.

Table 1: Datasets used in this research [4, 11]

Name	Questions	Answers	Domains
ASAP-sas	10	17,207	English, Biology, English Language Art, Science
Neural Network Course	17	646	Neural Network Course
SciEntsBank	331	10,804	12 domains
Beetle	47	366	Not known
Taxes-2011	80	2,273	Science

3.1.1 Data description.

ASAP-sas. The Automated Student Assessment Prize Short Answer Scoring (ASAP-SAS) dataset, introduced as part of a 2013 Kaggle competition sponsored by the Hewlett Foundation, comprises ten questions from four domains [1, 30, 34]. An answer typically contains 50 words, though a small fraction of answers (< 5%) exceed 100 words. Each question bears a score range of either 0 to 2 or 0 to 3. The competition’s structure and dataset promote the creation of real-world scoring systems. Its uniqueness lies in its varied question formats, including information-seeking prompts and image- or graph-based tasks requiring student interpretation.

Neural Network Course. The Neural Network Course dataset, available on GitHub, is sourced from a neural network course exam at the University of Applied Sciences Bonn-Rhein-Sieg. A single judge grades the exam using an integer scale: 0 entirely incorrect, one partially correct, and two perfect answers. This publicly available dataset has yet to be used in published research [14].

SciEntsBank. This dataset [23, 26, 31, 32] is a collection of student responses to questions from a standardized test administered to students in grades 3 to 6 across various schools in North America. SciEntsBank contains containing 12 domains. The grading requirement, contingent upon the domain category, encompasses a 2-way classification.

Beetle. The Beetle dataset [23] is created for a physics tutorial dialogue system targeting high school students, focusing on electricity and electronics. The dataset only includes relevant responses to factual or explanatory questions, disregarding interaction protocols.

Taxes-2011. The University of North Texas dataset [9, 21, 22] is derived from two examinations in Computer Science, with grades ranging from 0 to 5. Two evaluators grade each response without explicit rules, and their mean grade forms the ground truth, simulating real-world grading complexities.

3.1.2 Data splits. This research involves conducting experiments that demand different splits of the used datasets. The first experiments compare model performance on each dataset. For these experiments, each dataset is split into a training (70%), testing (20%), and validation (10%) set. The training and testing sets find application during the model training phase, reserving the validation set for estimating model performance after training completion. Subsequent steps involve concatenating the training, testing, and

validation sets from each dataset and subjecting the models to training and testing on these combined datasets. This ensures that every split has samples from each dataset and allows for evaluating various models’ performance on each individual dataset, facilitating comparison of model performance across different datasets and exploring whether certain models exhibit better compatibility with specific datasets. The final step entails training and testing the models on the concatenated datasets to estimate which model performs the best across multiple datasets.

This research also implements an alternate splitting method on the concatenated datasets by removing one of the datasets. The remaining datasets undergo an even split into a training and test set. The research uses the removed dataset as the validation set, providing a means to estimate model performance on new ASAG datasets. The model’s performance receives evaluation for every possible combination of a left-out dataset (cross-validation).

The concatenated dataset is also split based on the question domains. The questions of which the domains are unknown are filtered out. These experiments conduct the same splits as described above based on the datasets: within individual domains, within the concatenated domains dataset and by leaving one domain out of the training and test set to use as a validation set. The training, test and validation sets contain the same percentage of answers from their respective datasets when possible. The within domains split allows me to estimate the performance of different models for individual domains. This study uses the concatenated domains dataset to estimate model performance across domains. I use the split that leaves one domain out of the training and testing dataset to estimate the models’ ability to transfer knowledge across ASAG domains.

3.1.3 Sampling. To mimic limited training resources, the BERT model uses samples of the respective dataset for each experiment. The total sample size is 2,000. The experiments using individual datasets sample the dataset first and are then split into the train, test and validation set using the 70-20-10 splits. The experiments using the concatenated datasets start by taking out the validation dataset. The remaining concatenated set is sampled using equal proportions of each group (the data sets in the concatenated data set and the domains in the concatenated domains). This sampled dataset is split into train and test sets. The validation set is sampled to its appropriate sample size. The train set contains 1,400, the test set contains 400, and the validations set contains 200 student answers.

The model that uses shot learning to make a grade prediction is shown the examples from the training split of the related dataset. One example of each dataset is given when the model’s performance on the concatenated datasets is tested. Finally, for the domain split, at least one example (unless zero-shot learning) from each domain is included with the five examples given for every prediction. The examples are all chosen randomly from the related datasets for each prediction.

This study maintains the existing distributions of the ground truth categories in the datasets during sampling. This decision ensures that the model’s learning process and subsequent predictions reflect the real-world distributions of ASAG datasets and realistic performance comparisons with models trained on complete datasets.

3.2 Model implementation

Multiple variations of two fundamentally different approaches to building an ASAG model are being tested in this study. The first approach separates the process of embedding the student and reference answers from the model that grades the student's answer. The second approach integrates these two processes by not explicitly outputting the vectorized embedding of the student and reference answer but rather integrating the embedding and grading process into one model.

The following paragraphs describe the implementations of the embedding-based, BERT and LLM models. A more in-depth description of the Python libraries I use for each implementation can be found in the appendix section A.1.4, A.1.5, A.1.6, respectively.

3.2.1 Embedding-based models. BERT changes the vector representations of the words based on their context. Word2Vec does not do this and produces the same vectors for words, irrespective of their context. Having the vector representations change based on their context makes capturing more accurate representations of the words possible. However, they can also make it more difficult for the following models to train on. I use BERT, Fasttext, Conceptnet and Glove to embed the student and reference answers in this paper. A single vector of the student and reference answer is created by adding and averaging the word embeddings in the respective answers. All experiments are conducted using both the added and the averaged vectors.

The distance between the vector representations of the student and the reference answer is calculated using cosine similarity. Three regressions are used to predict the normalized points based on the cosine similarity: Isotonic regression, Linear regression and ridge regression (as used in [9]).

3.2.2 BERT. For the BERT model, the student and reference answers are tokenized using the `encode_plus` method from the AutoTokenizer inside the transformers library [35]. This method tokenizes the text for the BERT model and adds a token '[SEP]' between the student and reference answer, allowing the model to differentiate these inputs. I use the `BertForSequenceClassification` class from the transformers library by Hugging Face [35] for grading. This class consists of a BERT model with a sequence classification/regression head on top. The model is configured with `num_labels` equal to 1, which sets the model to compute a regression loss (Mean-Square loss). During training, the model aims to minimize the mean square error of the normalized assigned points and the predicted points. The BERT model is fine-tuned using only the training set, with all layers of the BERT model actively participating in the learning process (i.e., no layers are frozen). The model's performance is then evaluated on the separate test and validation set.

3.2.3 LLM. Zero, three and five-shot learning are used for the GPT-3.5turbo model, where examples are given of graded students and reference answers concerning the applicable dataset or domain for every API call. The following structure is sent to the API (a more detailed version can be found in the appendix A.1.6):

- instruction line (as an object):
 - “Grade the following Student answer based on the Reference answer. The grade should be a whole number.”
- the shots:

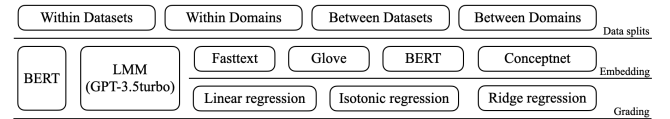
- “{instruction_line}
Student answer: {row['student_answer_i']}
Reference answer: {row['reference_answer_i']}
Grade out of
row['max_points']}: {row['assigned_points_i']}”
- prompt prediction:
 - “{instruction_line}
Student answer: {row['student_answer']}
Reference answer: {row['reference_answer']}
Grade out of
row['max_points']}: ”

The instruction_line inside the shots and the prompt is as follows: “Grade the following Student answer based on the Reference answer. The grade should be a howl number.”.

3.3 Experimental setup

Figure 1 gives an overview of the experiments conducted. The performance of the BERT model (as a grading model) is measured after five and ten training epochs. Each experiment uses the same training, test and validation split for every model training and performance measurement to ensure valid comparisons.

Figure 1: Experiments



3.4 Evaluation metrics

3.4.1 Evaluation metrics. All models except for the LLM model predict the normalized points values. This value is calculated based on the max points for each question and results in a continuous value between zero and one. This value is converted back into the originally scaled points based on the max points for each question. This allows these models to be trained across datasets with divergent max points. The RMSE and Pearson correlation are calculated using the originally scaled points. Using these original points to calculate these metrics allows for performance comparisons with articles using the same data sets. Each model's accuracy and weighted F-1 score is calculated by rounding the originally scaled points (using round half to even) to integers and using the model's predicted int grade and the assigned points grade as a class. The integers from zero to max points are the number of potential classes.

The LLM (GPT-3.5 turbo) model predicts a float value scaled to the original points. This is used because, after some short testing before running the experiments, the model performs substantially worse when predicting the normalized values. The performance metrics are calculated in the same way as the other models.

3.4.2 Weighted F1. The primary performance metric reported in this paper is the weighted F1 score. Other metrics with descriptions are available in the appendix A.1.7. The F1 score is a performance measure that considers both precision (the proportion of true positive results among all positive results predicted by the model) and

recall (the proportion of true positive results among all actual positive results). Given that the maximum points obtainable by students for a question range from one to five, there could be an imbalance in class distribution. To account for this, when aggregating the F1 scores into a single score, the weighted F1 score is employed. It is important to note that the weighted F1 score attributes more weight to classes with a higher number of instances, making it more sensitive to the model’s performance on majority classes.

3.4.3 Statistical tests. The following methods are used to determine if the performance of the datasets and models are significantly different. The Shapiro-Wilk test is first conducted to determine if the performance metrics follow a normal distribution, which is a prerequisite for using parametric tests. In each experiment conducted to discern significant differences in performance metrics, at least one contender does not appear to follow a normal distribution (Shapiro-Wilk test $p < 0.05$). Therefore, all statistics utilized in this study to detect performance differences are based on non-parametric methods, specifically the Kruskal-Wallis H test.

The Kruskal-Wallis H test is used to determine whether there are significant differences in the performance metrics among any of the tested samples. When a statistically significant Kruskal-Wallis H test result is obtained, Dunn’s test is subsequently employed to identify which specific samples differ significantly from each other. Dunn’s test, a post-hoc method, is used to identify pairwise differences between groups. It also adjusts the p-values to control the overall Type I error rate, mitigating the risk of false positives.

4 RESULTS

The results for the vast array of embedding-based models are reported solely in the appendix A.2. This is because their performance, as shown later in this section, does not significantly differ from each other and conserves space in the main body of the text.

The weighted F1 score for each experiment can be found in Figure 2. The grading models are a shortened concatenation of the different identifying aspects of the experiment (e.g. BERT_epochs_10 refers to bert-base-cased trained on ten epochs and BERT_avg_Isotonic refers to averaged BERT word embedding, graded using Isotonic regression). The datasets are either just named by the name of the dataset (if the models are trained and evaluated on one dataset) or made up of the concatenated domains or datasets and the name of the drop-out set. The ordering of the experiments corresponds to their average weighted F1 scores, with the top experiments having the highest scores (Figure 2).

Figure 3 visualizes the performance (found in Figure 2) of each model on the data splits: the individual datasets, the drop-out datasets (left_out_datasets in Figure A.2.1), the individual domains and the drop-out domains (left_out_domains in Figure A.2.1). The data behind this figure are found in the appendix A.2.1.

An overview of the metrics used to determine the significance of performance differences can be found in the methodology subsection: 3.4 and the interquartile ranges of the weighted F1 scores are reported in the appendix A.2.2.

4.1 LLM

Overall performance. This study uses zero, three, and five-shot learning to assess the performance of the LLM model (GPT-3.5turbo).

Figure 2: Weighted F1 scores

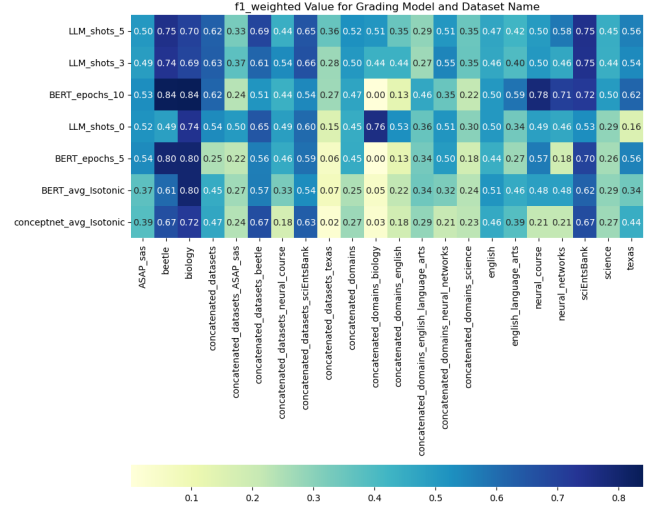
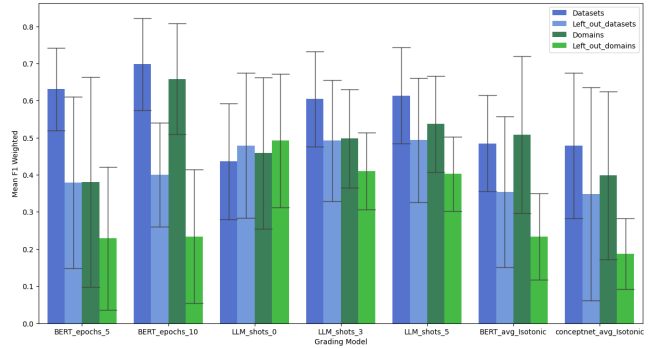


Figure 3: Mean and std weighted F1 across (unseen) datasets and (unseen) domains



The model’s performance varies with each condition. The zero-shot condition obtains an average weighted F1 score of approximately 0.468 (std = 0.161), signifying moderate variability in performance across the datasets. Under the three-shot learning condition, the model’s average performance improved to a weighted F1 score of 0.499, with slightly less variability (std = 0.135). The model achieves its best average performance in the five-shot learning condition with a weighted F1 score of 0.514 with slightly higher variability (std = 0.142). Overall, the variation in performance between datasets is relatively low for the LLM model compared to the other models, as shown in Figures 4.

Impact of shots on performance. The weighted F1 scores indicate an overall upward trend in the LLM model’s performance as the number of shots increases, signifying the value of providing more examples in the prompt. Nonetheless, the weighted F1 score with the zero, three and five shots do not significantly differ from each other ($p > 0.05$). The variation is higher for the five-shot performance than that of three shots, though this is primarily due to the increase in performance on a few datasets (Figure 2).

Performance across (unseen) datasets and (unseen) domains. Interestingly, the LLM model achieves a higher mean weighted F1 score on zero shots with the drop-out domains ($mean = 0.492, std = 0.180$) than when given three ($mean = 0.410, std = 0.103$) and five shots ($mean = 0.402, std = 0.100$) from other domains while achieving similar weighted F1 scores for the left out datasets (Figure 3). The superior performance of zero shots on unseen domains shows that giving examples from unrelated domains can diminish the performance of the LLM. Though, none of the shots variants show a statistically significant difference between the datasets and left_out_datasets nor the domains and left_out_domains ($p > 0.05$). This indicates consistent performance between datasets and unseen datasets, and domains and unseen domains by LLMs irrespective of the amount of examples given.

4.2 BERT

Overall performance. This study examines the performance of the BERT model after five and ten training epochs on a sample size of 1,400 (excluding 400 test samples and 200 validation samples) student answers from the respective datasets. After five training epochs, the BERT model achieves a mean weighted F1 score of 0.409 ($std = 0.236$). After ten training epochs, the mean weighted F1 score increases to 0.495 ($std = 0.231$), demonstrating an overall performance improvement compared to the five epochs model while maintaining a stable standard deviation. The models perform best on the beetle, biology, neural course (after ten epochs) and SciEntentsBank. These datasets, bar SciEntentsBank, are relatively small compared to the others (Figure 2).

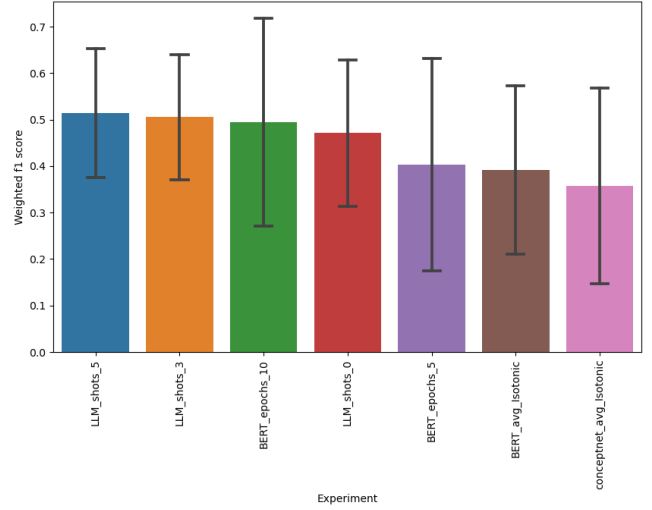
Impact of epochs on performance. Even though there is an observable difference in performance (Figure 4), the median F1 score of the BERT model after five epochs does not significantly differ from the BERT model after ten epochs ($p > 0.05$).

Performance across (unseen) datasets and (unseen) domains. Both after five and after ten epochs, the BERT model achieves a lower mean weighted F1 score on the unseen datasets (mean weighted F1 drop after five epochs: 0.252, and after ten: 0.298) and unseen domains (mean weighted F1 drop after five epochs: 0.131, and after ten: 0.424) compared to its performance when trained and tested on the same datasets and domains. This drop is only statistically significant after ten training epochs ($p < 0.05$). Interestingly, the increase in performance after ten epochs compared to five is solely due to the increase in the non-drop-out datasets and domains. The mean weighted F1 score on the drop-out domains and datasets barely differ after five and ten training epochs (Figure 3). This suggests that in the initial five epochs of training on the concatenated datasets (excluding the drop-out dataset) or concatenated domains (excluding the drop-out domain), the BERT model seems to reach its optimal capacity for grading the drop-out dataset.

4.3 Embedding-based models

The results section only reports on the results of Conceptnet (embedding) using isotonic regression (grading) and BERT (embedding) with Isotonic regression (grading) because these models achieve the highest mean weighted F1 scores. The F1 scores of all experiments can be found in appendix A.2.

Figure 4: The mean and std weighted F1 weighted values for best performing grading models



Overall performance. Based on the average F1 score, the strongest-performing embedding models are BERT and Conceptnet. Averaging or adding up the word embeddings within the student and reference answer did not affect the results. Also, no significant difference in performance is found between the different grading models (Ridge, Linear, and Isotonic Regression) using the separated embeddings ($p > 0.05$). The BERT and Fasttext are the only models that perform significantly differently from each other ($p < 0.05$).

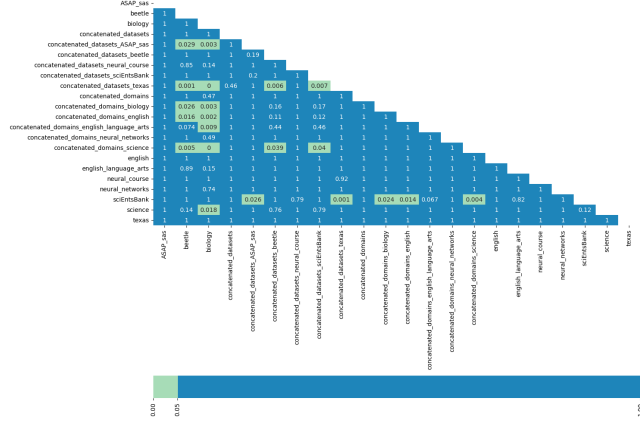
Performance across (unseen) datasets and (unseen) domains. As can be observed in Figure 3, the embedding-based models achieve lower mean weighted F1 scores for the left_out_datasets and left_out_domains compared to their non-drop-out counterparts. Nonetheless, this drop is not significant in either case ($p > 0.05$).

4.3.1 Comparisons.

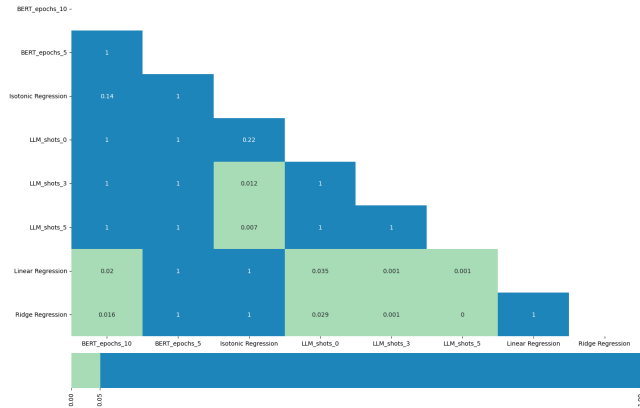
BERT vs LLM. The LLM model, using three and five-shot learning, reaches a higher average F1 score than the BERT base model trained on the 1,400 student answers sample, and its performance is more consistent throughout the individual datasets, the drop-out datasets and the drop-out domains (Figure 4 and 3). The BERT model does, however, reach higher F1 scores on specific datasets: beetle 0.84 (BERT 10 epochs) vs 0.75 (LLM 5 shots), biology 0.84 vs 0.7, neural course 0.78 vs 0.5, neural networks 0.71 vs 0.58 (Figure 2).

Datasets. The performance of the different models on the datasets is only significantly different for a few dataset pairs ($p < 0.05$), as can be observed in Figure 5. The models do not perform significantly differently on most datasets experimented on. The two datasets with the lowest p-value in the Dunn's test are concatenated_domains and concatenated_datasets_left_out_texas ($p < 0.01$).

Statistical Comparisons of Grading Models. Statistical analysis of the grading models (Isotonic Regression, Linear Regression, Ridge Regression, bert-base-cased_epochs_5, bert-base-cased_epochs_10,

Figure 5: Dunn’s Post-hoc Test of the weighted F1 scores between datasets

LLM_shots_0, LLM_shots_3, LLM_shots_5) revealed significant differences in the weighted F1 scores between at least two models ($p < 0.05$). A post hoc Dunn’s test indicates no significant difference in the median scores among the regression models ($p > 0.05$). Likewise, the scores obtained by BERT and the LLM were not significantly different ($p > 0.05$), and bert-base-cased_epochs_10 presented a significant difference only when compared with Ridge Regression and Linear Regression ($p < 0.05$), but not with Isotonic Regression or other grading models. LLM_shots_0 show a significant difference only compared to Ridge Regression ($p < 0.05$). In contrast, LLM_shots_3 significantly outperformed all regression models, while LLM_shots_5 demonstrated substantial superiority over the regression grading models ($p < 0.01$) but no significant difference from the GPT and BERT grading models ($p > 0.05$). Figure 6 presents these results in more detail.

Figure 6: Dunn’s Post-hoc Test of the weighted F1 scores between grading models

5 DISCUSSION

5.1 Grounding performance

To ground the results obtained in this study and to assess the extent to which the performance of each model on the diverse data sets is representative, I compare my findings to those from other studies.

beetle. The beetle dataset has been used in one other study [23] that uses a combination of handcrafted features to grade the student answers. This model achieves around 75% accuracy comparable to the LLM model with three (accuracy: 73.87%) and five shots (accuracy: 75.08%).

SciEntsBank. Early approaches, using hand crafted features, achieve around 70% accuracy [23, 31] on the SciEntsBank dataset. In later studies, using BERT, the accuracy increases to 75.9% (weighted F1: 75.8) [32] while another study only reaches 72.8% (weighted F1: 72.8) using the same model [26]. The GPT-3.5turbo with three and five shots both achieve a weighted F1 score of 0.75 (also 75% accuracy on both). While the BERT model trained for this study, using 2,000 samples and training on five and ten epochs, achieves 0.70 and 0.72 weighted F1 scores, respectively (both have the same accuracy).

Texas. The performance metrics used on the Texas dataset [21, 22] are RMSE and Pearson correlation. An early corpus-based approach, using a similarity score to assign points, achieves 0.37 Pearson correlation [22]. BERT embeddings, cosine similarity, and Isotonic regression achieve a similar Pearson correlation of 0.334. This increases to 0.417 when using Conceptnet. An approach using handcrafted features achieves a Pearson correlation of 0.570 and RMSE of 0.902 [31]. A third article uses FastText, Glove, BERT, GPT-2 and a few models to embed the student and reference answers and achieves Pearson correlations of 0.537, 0.509, 0.318, and 0.248, respectively. The accompanying RMSE are 0.956 for FastText, 1.002 for Glove, 1.065 for GPT-2, and 1.057 for BERT [9]. In this study, the LLM_shots_3 achieves a Pearson correlation of 0.592 and an RMSE of 1.120. The BERT model achieves 0.562 after five epochs and 0.625 Pearson Correlation after ten epochs, and 0.839 and 0.814 RMSE, respectively, outperforming the earlier models. The models embedding-based models in this study align with the performance of similar models in papers. As expected, the models in this article and earlier articles, both using BERT embeddings, are very close, 0.318 [9] and 0.334 RMSE.

ASAP-SAS. Using a traditional embedding-based model, 0.7439 [30] and 0.773 [34] mean Quadratic Weighted Kappa (mean QWK) is achieved on the ASAP-sas data set. Using a similar approach, though now using a BLSTM to grade the embeddings, a 4.79 RMSE and 0.71 Pearson correlation are reached [1]. This study finds that the BERT_avg_Isotonic Regression achieves a 0.765 RMSE and 0.379 Pearson correlation. Bert-base-cased_epochs_5 (RMSE: 0.759, Pearson Correlation: 0.575), bert-base-cased_epochs_10 (RMSE: 0.724, Pearson Correlation: 0.620), LLM_shots_0 (RMSE: 0.768, Pearson Correlation: 0.610), LLM_shots_3 (RMSE: 0.922, Pearson Correlation: 0.529), LLM_shots_5 (RMSE: 0.768, Pearson Correlation: 0.610) achieve substantially lower RMSE and Pearson correlation. Given that RMSE

should be lower and Pears Correlation higher for better performance, every model in this study interestingly obtains a much lower RMSE (better) but a weaker Pearse Correlation.

Neural course. The Neural Network Course datasets [14] has not yet been used in published research. Therefore, the results obtained in this study cannot be grounded in earlier research for this dataset.

The models used for this paper perform comparably to the models observed in earlier articles on the same datasets. By adding these comparisons to earlier works, this article hopes to indicate how these different approaches compare to each other and to extend an estimate of how they would perform on new datasets and domains.

5.2 LLM

The LLM model performs consistently over the different datasets compared to the other models and achieves the highest average weighted F1 score with three and five-shot learning. This shows the strong potential of these models in the ASAG field. Given that increasing the shots indicates a clear, albeit not significant, upwards trend in performance, future research could investigate this trend in more detail, plotting the performance trajectory as the number of shots increases. Furthermore, there is potential to experiment extensively with diverse shots to investigate their impact on performance.

Interestingly, the zero-shot approach performs comparably on the drop-out datasets and better on the drop-out domains, albeit not significantly. This indicates that examples of unrelated checked student answers can diminish the performance of LLMs in ASAG. Future research could map out the clusters of domains that enhance performance for each other. The consistent performance of the zero-shot LLM also shows that they can be good starting models for new ASAG datasets or domains.

5.3 BERT

The results show that the BERT model performs very well on relatively small datasets. A viable explanation for BERT's enhanced performance on these datasets could be overfitting to the questions, a potential consequence of their limited size. However, if the variety and nature of exams and student responses remain fairly stable, it is reasonable to expect that BERT might continue to achieve high accuracy when grading new responses from the same exams.

5.4 Embedding based models

The embedding-based models do not differ significantly in performance. Neither the type of regression to grade the student answer embedding nor the embedding model significantly impact the performance. This indicates that BERT contextual vector representations do not impact grading performance. One explanation could be that the model used for grading cannot pick up on the additional features embedded by the BERT model.

5.5 Comparisons

As observed in the results, BERT achieves higher performance on specific datasets, but its performance drops significantly on the drop-out datasets and domains. In contrast, the LLM performs more consistently across the datasets, drop-out datasets, domains and

drop-out domains. This is likely because the LLM has an even more robust understanding of language and thus has more background knowledge to draw from when grading. Despite this, it may struggle to achieve optimal accuracy as it may not have sufficient exposure to the specific features necessary for accurately grading a particular dataset. This highlights the potential limitation of relying solely on few-shot learning without considering dataset-specific factors.

Embedding-based models underperform in comparison to both BERT and the LLM. Despite this, the variance in performance between the embedding-based models and BERT is not substantial. This implies that while BERT is proficient at assimilating grading features unique to specific data sets, it encounters difficulties when applying its language understanding skills to grade novel data sets.

6 CONCLUSION

This study makes three significant contributions to the field of Automated Short Answer Grading (ASAG). Firstly, it provides a comprehensive analysis of the performance of the BERT model across diverse datasets and domains, employing cross-validation to evaluate its performance on unseen data. Secondly, it uses diverse performance metrics to investigate the efficacy of traditional grading models, where the embedding and grading are performed by separate models, the current state-of-the-art (BERT) and a new LLM model (GPT-3.5turbo). The utilization of various performance metrics across multiple datasets provides a detailed account of the relative strengths and weaknesses of these models and a perspective on the relative difficulty of individual datasets. Finally, this article shows the potential of the use of LLMs in the field of ASAG and gives a comprehensive overview of its performance.

BERT outperforms the LLM on individual data sets, particularly smaller ones. This outcome underlines BERT's ability to discern grading attributes specific to individual contexts. However, this could indicate potential overfitting, a hypothesis that requires further examination. The BERT model's performance significantly decreases on dropout datasets and domains, suggesting a limited ability for the model to transfer ASAG knowledge across different domains and datasets.

The LLM model demonstrates consistent, strong performance across various datasets and domains, indicative of its robustness and generalization capabilities. This underscores the value of LLMs within ASAG. Notably, our experiments revealed that the LLM, employing three and five-shot learning, surpasses the performance of the BERT model, trained on samples of 2,000 student responses over ten epochs when measured using the average weighted F1 score across all experiments.

The reliably strong performance exhibited by the LLM in the context of few-shot learning invites further exploration. Research to map the trajectory of its performance with changes in the number of shots could yield insightful results. The potential for experimenting with various prompts for LLM models in ASAG also merits further investigation.

This study demonstrates the potential utility of LLMs as an initial point of reference for new ASAG tasks. The promising results point to the progressive advancement of the ASAG field, stimulating further research dedicated to enhancing the application of LLMs in ASAG.

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A APPENDIX

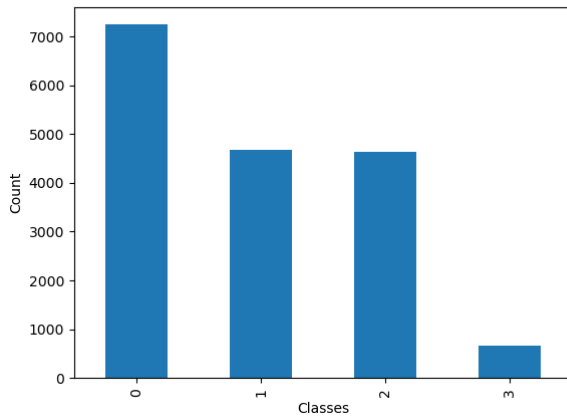
A.1 methodology

A.1.1 Data.

ASAP-sas. The Automated Student Assessment Prize Short Answer Scoring (ASAP-SAS) dataset, introduced as part of a 2013 Kaggle competition sponsored by the Hewlett Foundation, comprises ten questions from four domains 1 [1, 30, 34]. The training data set contains 17,207 answers (about 1,700 per question), and

the test set contains 5,224 answers. This study only uses the publicly available training set. An answer typically contains 50 words, though a small fraction of answers (< 5%) exceed 100 words. Each question bears a score range of either 0 to 2 or 0 to 3. The competition and the dataset are structured to confront assessment systems with practical and diverse questions, promoting the development of scoring systems for real-world educational applications due to the variety of the ten questions. A distinctive attribute of this dataset is the broad range in question structure, with some questions seeking specific information from a one- or two-page prompt text. In contrast, others include images or graphs and require students to record their observations and interpretations of the results. Figure 7 shows the distribution of assigned points inside the ASAP-sas dataset. The category of three assigned points is the most underrepresented class in this dataset.

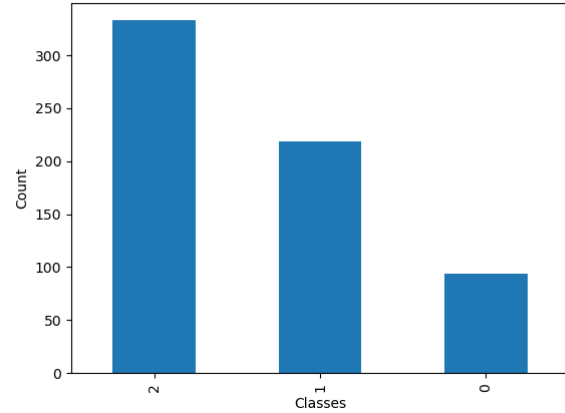
Figure 7: ASAP-sas assigned points distribution



Neural Network Course. The Neural Network Course dataset is available on GitHub. It is sourced from a neural network course administered at the University of Applied Sciences Bonn-Rhein-Sieg. It is comprised of student responses to exam questions collected. A total of 38 students took the examination, each answering 17 questions, thereby generating a dataset of 646 individual responses. Each of these responses is graded by a single judge on an integer scale: 0 indicates an entirely incorrect answer, 1 indicates a partially correct answer, and 2 indicates a perfect answer. This publicly available dataset has yet to be used in published research [14]. The zero points category is the least represented, as shown in Figure 8. It is about half the size of the one-point category and one-third the size of the two points category.

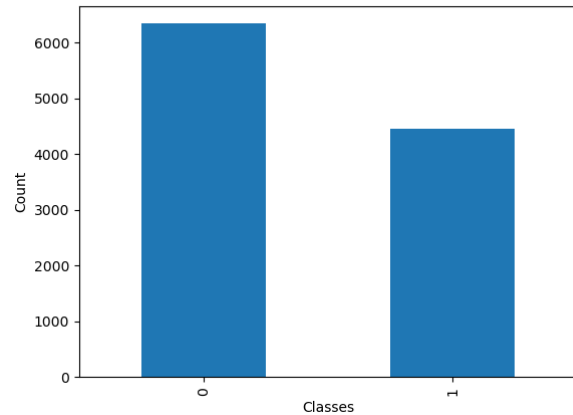
SciEntsBank. This dataset [23, 26, 31, 32] is a collection of student responses to questions from a standardized test administered to students in grades 3 to 6 across various schools in North America. SciEntsBank contains 10,804 responses to 331 questions, containing 12 domains. The grading requirement, contingent upon the domain

Figure 8: Neural Networks Course assigned points distribution

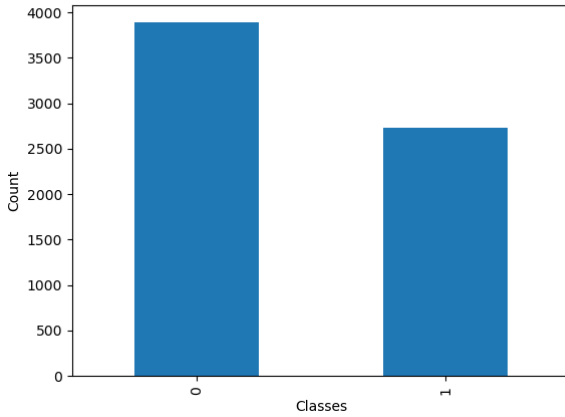


category, encompasses a 2-way classification. The dataset has relatively equal data distribution, with about 4,500 one assigned points class samples and just over 6,000 zero points class samples 9.

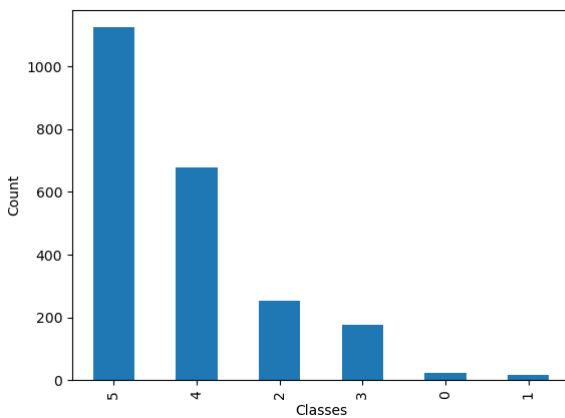
Figure 9: SciEntsBank assigned points distribution



Beetle. The Beetle dataset [23] is designed to evaluate student engagement with a tutorial dialogue system. This system is geared towards educating high school students in the principles of physics, with a specific focus on electricity and electronics. The dataset has been derived by revising dialogues and selecting only pertinent responses to questions, disregarding the interaction protocol. The types of questions posed are either factual or involve explanation and definition. The corpus includes 47 questions, averaging 366 student responses per question. As can be observed in Figure 10, there are more examples of the zero assigned points class, though there is a substantial amount of both zero (almost 4,000) and one (just above 2,600) assigned points.

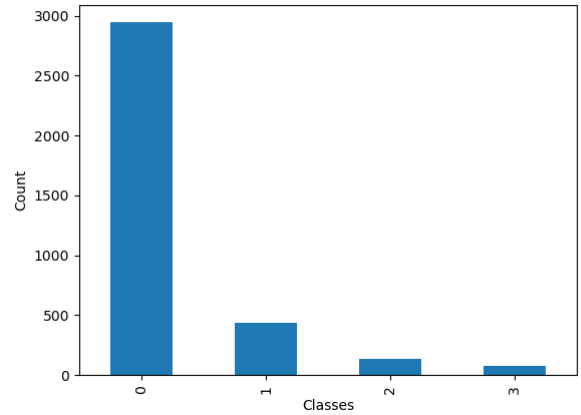
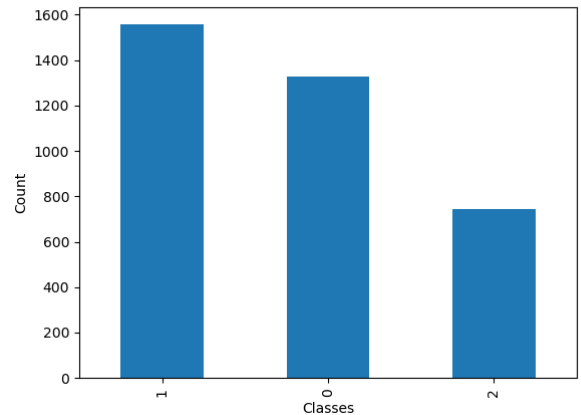
Figure 10: Beetle assigned points distribution

Taxes-2011. The University of North Texas dataset [9, 21, 22] contains responses from 30 students to 80 varied questions. This results in approximately 2,300 question-answer pairings, with each answer averaging 50-word tokens in length. The questions originate from 10 assignments within two examinations assessing fundamental Computer Science knowledge. Each response comes with a grade ranging from 0 to 5, assigned by two evaluators. The grading system employs integer numbers. The grading process does not follow any explicit rules, and the mean grade provided by the two evaluators is treated as the ground truth. This dataset aims to replicate the real-world challenges of grading short answers. The majority of assigned points in the Taxes-2011 datasets are four and five. A very low amount of zero and one assigned points class samples are present in the dataset (Figure 11).

Figure 11: Taxes-2011 assigned points distribution

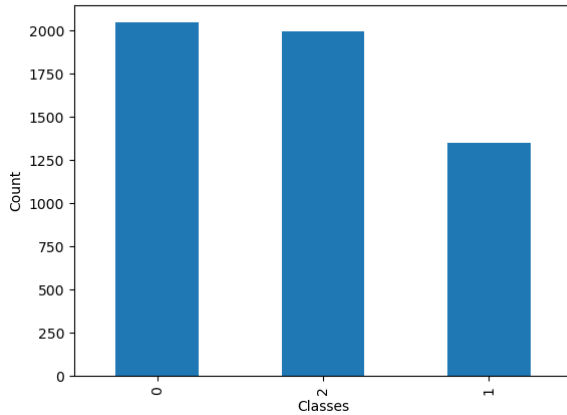
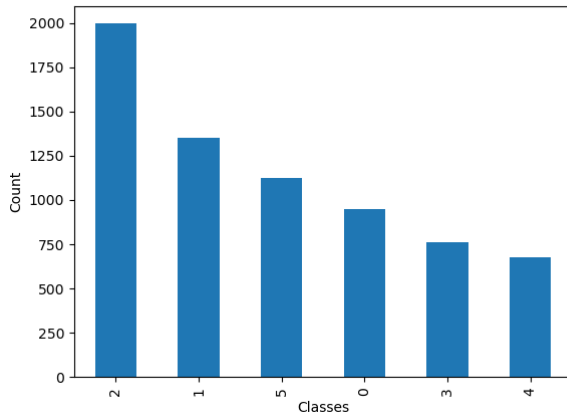
The following figures show the distribution of the assigned points of the remaining datasets. These datasets are subsets or

concatenations of the above-described datasets and are thus not described in depth.

Figure 12: Biology assigned points distribution**Figure 13: English Language Arts assigned points distribution**

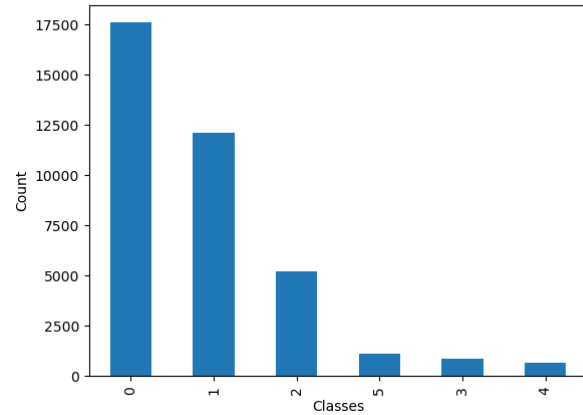
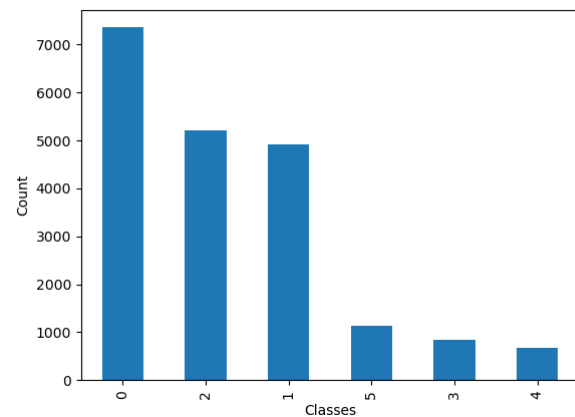
A.1.2 Standardization and Cleaning. The structure of each dataset is standardized by renaming and dropping columns. This study standardizes the raw datasets into the following columns: 'row_id', 'question_id', 'student_answer', 'assigned_points', 'question', 'reference_answer', 'domain', 'max_points', 'dataset_name', 'normalized_points'. I convert missing student answers into empty strings.

The Beetle and SciEntsBank datasets use boolean values to grade the student answers. The boolean values are converted into 0 (incorrect) and 1 (correct). The datasets ASAP-sas and Texas-2011 have two graders. The average points given by the graders are used as the ground truth and rounded using round half to even. The points in all the datasets are normalized to values between zero and one based on the maximum obtainable points for each question.

Figure 14: English assigned points distribution**Figure 15: Science assigned points distribution**

The neural course does not contain the maximum available points. In this case, the maximum points assigned for each question are used as the maximum obtainable points. This assumes that at least one student's answer received full points for every question in the dataset. Only the student answers contain null values in some datasets; these are converted to an empty strings.

The ASAG-sas dataset available online does not include reference answers. The reference answers are created using the following methods. The student answers that did not receive full points are filtered out of the dataset. The correct student answers are embedded using TfidfVectorizer, and for each question, an average vector value is calculated based on all the correct student answers related to the question. The distance between the vector of each correct student answer and the average answer vector of the related question is then calculated. For each question, the correct student answer that is closest to its average correct student answer is selected as the reference answer. The text of this reference answer is

Figure 16: Concatenated datasets assigned points distribution**Figure 17: Concatenated Domains assigned points distribution**

added as the reference answer for each question in the ASAG-sas dataset.

A.1.3 Text data model preparation. The text processing is altered for each model to best represent the pre-training data. A detailed description of the text processing conducted for each model is found below.

Conceptnet, Fasttext, Glove. The following pre-processing is conducted on both the student and the reference answers: All text is lowered, all non-text items are removed, extra whitespace is stripped, the spelling is corrected (using SpellChecker from the spellchecker python library), punctuation is removed, nltk.word tokenize is used to tokenize the student and reference answer (these models are trained on text with this tokenization method), stopwords are removed (using the nltk stopwords library), and the tokens are lemmatized.

BERT. The following pre-processing is conducted on both the student and the reference answers: the spelling is corrected (using SpellChecker from the spellchecker python library), AutoTokenizer from the transformers [35] python library is used for tokenization, the token [SEP] is used to differentiate the student and the reference answer, an attention mask is added to indicate the tokens that represent the text and the tokens that fill the input window when the tokenized student answer and reference answer are shorter than 512, the tokens are truncated if they exceed BERT’s max token length of 512. This pre-processing aligns with the pre-processing of the pre-training data of the bert-base-cased model used for this study [40].

GPT-3.5turbo. The GPT-3.5turbo model uses the raw text of the student and reference answers. This is in line with the pre-training data [5].

A.1.4 Embedding-Based Models. Using its downloader API, I utilize the pre-trained FastText, GloVe, and ConceptNet models from the Python package Gensim [29]. For BERT word embeddings, I utilized the BertModel and BertTokenizer classes from the transformers library and the pre-trained BERT model ‘bert-base-uncased’ [35]. To calculate the cosine similarity between the student and reference answer embeddings, I use the cosine function from the scipy.spatial.distance module [33]. The Isotonic, Linear, and ridge regression are implemented using the sklearn python library [24].

A.1.5 BERT. The method `get_linear_schedule_with_warmup` from the transformer library is used to create a learning rate scheduler for the optimizer. The `num_warmup_steps` is set to zero, and the `num_training_steps` is set to ten. The AdamW optimizer is used with a learning rate of $2e-5$ [35].

A.1.6 LLM.

- instruction:
 - “role”: “system”
 - “content”: “You are an AI trained to grade student answers based on a reference answer. Please return a single howl number.”
- the shots:
 - “role”: “system”
 - “content”: “{instruction_line}
Student answer: {row[‘student_answer_i’]}
Reference answer: {row[‘reference_answer_i’]}
Grade out of {row[‘max_points’]}; {row[‘assigned_points_i’]}”
- prompt prediction:
 - “role”: “system”
 - “content”: “{instruction_line}
Student answer: {row[‘student_answer’]}
Reference answer: {row[‘reference_answer’]}
Grade out of {row[‘max_points’]}; ”

The `instruction_line` inside the shots and the prompt is as follows: “Grade the following Student answer based on the Reference answer. The grade should be a howl number.”

A.1.7 Evaluation metrics.

Recall Micro and Macro. Recall (or sensitivity) is a measure of a model’s completeness. It quantifies the number of correct positive

Figure 18: Table: Mean and std weighted F1 across (unseen) datasets and (unseen) domains

	mean Datasets	mean Domains	mean Left_out_datasets	mean Left_out_domains	std Datasets	std Domains	std Left_out_datasets	std Left_out_domains
BERT_epochs_5	0.631	0.38	0.379	0.229	0.111	0.283	0.231	0.192
BERT_epochs_10	0.698	0.658	0.4	0.234	0.124	0.149	0.14	0.18
LLM_shots_0	0.436	0.459	0.479	0.492	0.157	0.204	0.195	0.18
LLM_shots_3	0.604	0.498	0.492	0.41	0.128	0.133	0.164	0.103
LLM_shots_5	0.614	0.537	0.493	0.402	0.13	0.129	0.168	0.1
BERT_avg_isotonic	0.485	0.508	0.354	0.234	0.13	0.212	0.203	0.117
conceptnet_avg_isotonic	0.478	0.398	0.348	0.187	0.196	0.226	0.287	0.095

predictions made out of the actual positive instances. Micro Recall calculates the total true positives against the sum of total true positives and total false negatives across all classes. Macro Recall calculates recall for each class independently and then takes the average, which gives equal weight to all classes regardless of their size.

Precision Micro and Macro. Precision measures the model’s accuracy of positive predictions. It quantifies the number of correct positive predictions made. Micro Precision calculates the total true positives against the sum of total true positives and total false positives across all classes, essentially weighting each instance equally. Macro Precision calculates precision for each class independently and then takes the average of these precision scores, treating all classes as equally important.

F1 Micro and Macro. The F1 score measures a model’s accuracy, considering precision and recall. It is the harmonic mean of these two metrics. The F1 score reaches its best value at one and its worst at zero. Micro F1 score calculates the performance by considering the total true positives, false negatives, and false positives across all classes, treating all instances equally regardless of their class. This measure is often used when class imbalance is present. The Macro F1 score calculates the performance for each class independently and then takes the average of the scores, treating all classes equally important regardless of their proportion in the dataset.

Accuracy. Accuracy is the ratio of correct predictions to total predictions. Accuracy works well when the datasets are symmetric, i.e., when the classes are almost equally split. However, it can be misleading if there is a class imbalance.

RMSE and Pearson Correlation. Root Mean Square Error (RMSE) tells us how concentrated the predicted values are around the ground truth. It does this by squaring the differences between predicted and actual values, taking their average, and then square-rooting the result. A lower RMSE signifies a better performance. Pearson correlation coefficient is a measure of the linear correlation between two variables. It has a value between +1 and -1, where +1 indicates total positive linear correlation, 0 indicates no linear correlation, and -1 indicates total negative linear correlation. It measures how closely the predicted and actual values are correlated.

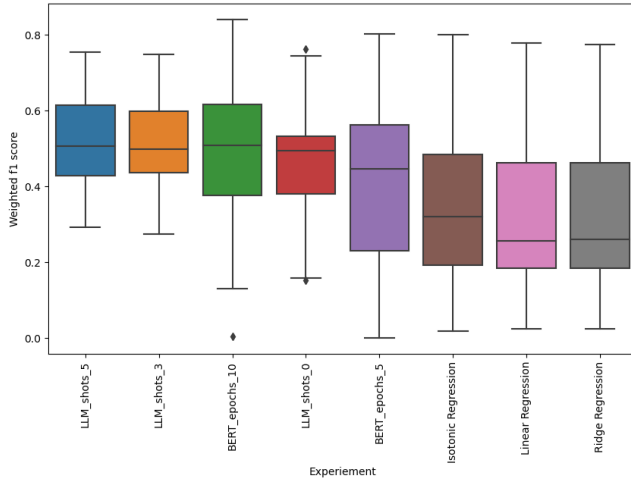
A.2 Results

A.2.1 Weighted F1 different dataset splits. Table 18 shows the data behind Figure 3 in the results section 4.

Figure 19: Weighted F1 score interquartile range

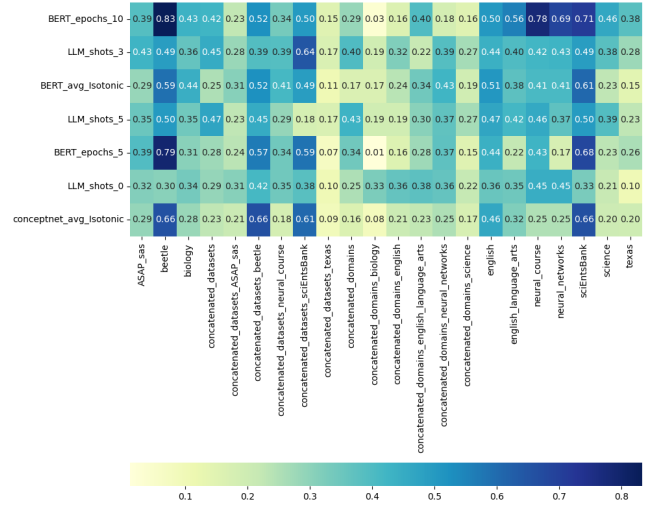
	min	25%	50%	75%	max	IQR
BERT_epochs_10	0.004	0.375	0.509	0.617	0.84	0.241
BERT_epochs_5	0.001	0.229	0.447	0.561	0.801	0.332
Isotonic Regression	0.018	0.193	0.319	0.484	0.799	0.291
LLM_shots_0	0.153	0.38	0.495	0.531	0.762	0.151
LLM_shots_3	0.273	0.435	0.498	0.598	0.747	0.163
LLM_shots_5	0.291	0.427	0.506	0.613	0.754	0.186
Linear Regression	0.024	0.185	0.256	0.462	0.778	0.277
Ridge Regression	0.024	0.185	0.259	0.462	0.773	0.277

A.2.2 Interquartile Range Weighted F1. Figure 19 shows the interquartile range of the weighted F1 score for each model visualized for each model in Figure 20. 4.

Figure 20: Weighted F1 interquartile ranges for grading model and dataset name, ordered by average weighted F1 value

A.2.3 Weighted F1, macro F1, micro F1 and accuracy. Figure 24 presents the obtained weighted F1 scores for all models tested. The subsequent three figures illustrate the macro F1 score (Figure 21), micro F1 score (Figure 22), and the accuracy (Figure 23), only including the best performing embedding based models (BERT_avg_isotonic and conceptnet_avg_isotonic). The micro F1 score and the accuracy align with the weighted F1 score, indicating the model's consistent performance across multiple metrics. The macro F1 score, in contrast, is substantially lower for all models, suggesting a significant disparity in performance across various classes.

The datasets that exhibit the least decline in macro F1 scores — English, Neural Course, Beetle, and ScieEtsBank — share a relatively equal distribution of assigned points. This suggests that the dip in Macro F1 score is likely due to the imbalanced class distributions,

Figure 21: Macro F1 score for grading models on datasets, ordered by average weighted F1 value

indicating that a scarcity of examples for all potential obtainable grades adversely affects all models.

These findings hint at the potential for marked improvements across all models in predicting underrepresented "assigned points" classes using samples or shots that use equal distributions of these classes. Such a yet-to-be-explored strategy within ASAG involves the undersampling of the majority classes and oversampling of the minority classes to reshape the assigned points' distribution in ASAG training datasets. While this strategy is beyond the scope of this article, future research should investigate the impact of this approach on model performance.

Figure 22: Micro F1 score for grading models on datasets, ordered by average weighted F1 value

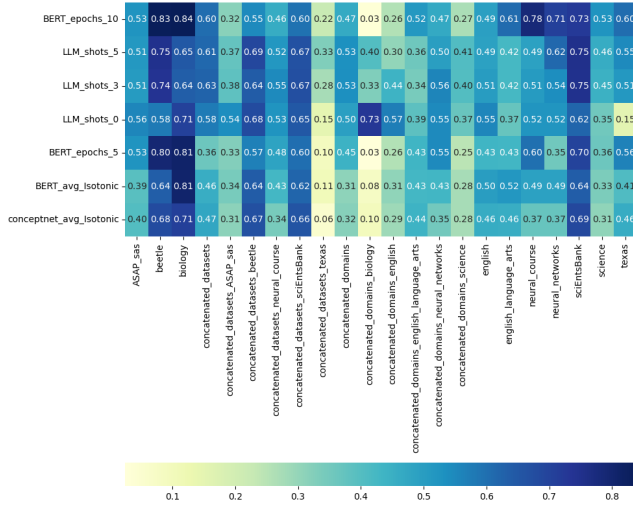


Figure 23: Accuracy of grading models on datasets, ordered by average weighted F1 value

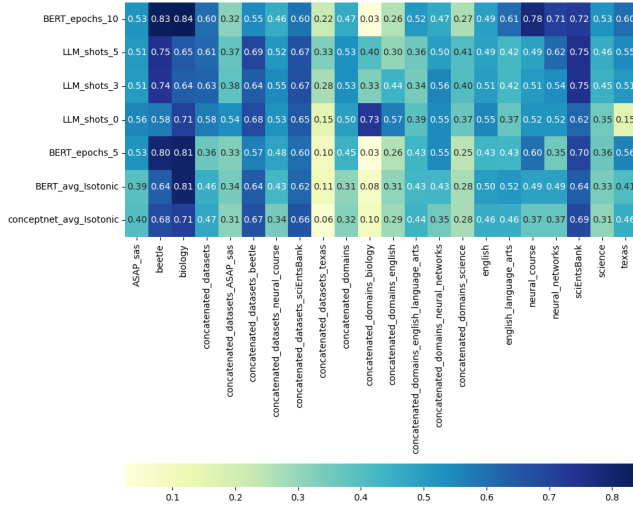


Figure 24: Weighted F1 score for all grading models on datasets, ordered by average weighted F1 value

